

DANYAAL KALEEL

Email: danyaal.kaleel@outlook.com

Robotics Engineer focused on developing next-generation buoyancy-driven soft robots. Proven track record of translating complex research into functional prototypes through rigorous design, construction, and laboratory testing. Passionate about advancing the frontiers of robotics and looking to contribute technical expertise to a research team.

EDUCATION

PhD, Queen Mary University of London, Mechanical Eng. Sep 2021 – Jul. 2026

- Thesis Titled: Buoyancy Actuation Using Incompressible Fluids in Soft Robots for Deep Sea Operations
- PhD Project: Designed, manufactured and experimentally validated buoyancy actuated soft robot arms for underwater operations.
- Skills: Public speaking, programming (Python, MATLAB, Arduino), manufacturing soft robots (ultrasonic welding, vacuum impulse welding, 3D printing), designing and executing experimental test plans.

BEng, Queen Mary University of London, Robotics Eng. Sep. 2018 – Jul. 2021

- Degree Classification: BEng First Class Honours
- Final Year Project: Developed an algorithm to navigate a wheeled mobile robot around a hospital wardroom for autonomous cleaning and disinfection. Algorithm was built in C++ and used ROS for controlling the robot. The algorithm was tested in the Gazebo robot simulator.
- Skills: Programming (Java, C, MATLAB, 8051 assembly language, ROS), Solidworks, Abaqus, Star CCM+.

WORK HISTORY

Module Demonstrator Sep. 2022 – Dec. 2024

School of Engineering and Materials Science and School of Electronic Engineering and Computer Science

- Organised workshops to enhance student's practical skills and understanding.
- Collaborated with academic staff to improve course delivery and content relevancy.
- Guided students through hands-on activities, fostering their problem-solving skills and creativity.
- Delivered lectures on robot maths to groups of 30 students.
- Provided technical support during laboratory sessions, mitigating potential disruptions in the learning process.
- Verbally explained complex concepts with complementing visual aids, enhancing clarity and retention rates among learners.
- Ensured safety whilst conducting demonstrations with adherence to strict safety protocols.

Engineering Ambassador Nov. 2023 – Jul. 2024

Proud to be an Engineer (Royal Academy of Engineering)

- Designed and delivered public engagement events to raise awareness of engineers from non-traditional backgrounds.

- Secured generous funding from Royal Academy of Engineering for the initiative.

Research Assistant

Jan. 2022 – Mar. 2022

Queen Mary University of London

- Collaborated with a team of engineers to design and build an eversion robot system capable of surveying the internal structure of buildings and applying insulating material.

SCHOLARSHIPS AND AWARDS

Award: Associate Fellowship of the Higher Education Academy (AFHEA) Nov. 2025
Issued by: Advance HE

Scholarship: PhD Studentship Sep. 2021
Issued by: Queen Mary University of London

Award: School of Engineering and Materials Science School Prize for Robotics for the Best Robotics Engineering Graduate Jul. 2021
Issued by: Queen Mary University of London

Award: Fred and Rose Zappert Prize for Outstanding Academic Achievement in the 2020/21 Academic Year Jul. 2021
Issued by: Queen Mary University of London

PUBLICATIONS

1. **D. Kaleel**, T. Lalitharatne, A. Baker, H. Wurdemann, B. Clement and K. Althoefer, "Buoyancy Actuation of an Underwater Soft Robot Arm through Inflation with Incompressible Fluids," Ocean Engineering (Under Review)
2. **D. Kaleel**, B. Clement, and K. Althoefer, "Modelling of Buoyancy Based Actuation of an Inflatable Underwater Soft Robot," TAROS (Towards Autonomous Robotic Systems), 2024.
3. **D. Kaleel**, B. Clement, and K. Althoefer, "Hydraulic Volumetric Soft Everting Vine Robot Steering Mechanism for Underwater Exploration, " IFAC-PapersOnLine, 2024.
4. C. Suulker, S. Skach, **D. Kaleel**, T. Abrar, Z. Murtaza, D. Suulker, and K. Althoefer, "Soft Cap for Vine Robots," 2023 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS), Detroit, MI, USA, 2023.
5. **D. Kaleel**, B. Clement and K. Althoefer, "A Framework to Design and Build a Height Controllable Eversion Robot," 2023 11th International Conference on Control, Mechatronics and Automation (ICCMA), Grimstad, Norway, 2023.

SKILLS AND INTERESTS

IT: Proficient in using Office365 suite, Python, C, MATLAB, Arduino.

Music: Achieved Grade 8 Distinction in Classical guitar.

Sport: Regular training at gym, five times a week.

Travel: Extensively travelled around Europe, Asia, Australia and America.